MIL-C-1470F
7 August 1972
SUPERSEDING
MIL-C-1470E
6 November 1967

MILITARY SPECIFICATION

CEREALS, BREAKFAST, PREPARED (READY-TO-EAT) AND UNCOOKED:

PACKAGING, PACKING, LABELING, AND MARKING OF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the packaging, packing, labeling, and marking requirements for the domestic and overseas shipment of all prepared (ready-to-eat) and uncooked cereal subsistence products for use by the Armed Forces as an item of general issue.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

N-C-195	-	Cereal, Rolled Oats.
N-C-196	-	Cereals, Breakfast, Prepared (Ready-To-Eat).
N-C-201	-	Cereals, Wheat; Uncooked.
N-C-521	-	Corn Meal (Regular and Degermed).
PPP-B-636	-	Boxes, Shipping, Fiberboard.

MILITARY

MIL-A-148	-	Aluminum Foil.
MIL-L-10547	-	Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.
MIL-L-35078	-	Loads Unit; Preparation of Nonperishable Subsistence.

FSC 8920

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STANDARDS

FEDERAL

FED-STD-101 - Preservation, Packaging, and Packing Materials; Test Procedures.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents forms a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

U. S. Department of Health, Education and Welfare

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, D. C. 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

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3. REQUIREMENTS

3.1 Materials.- All packaging and packing materials shall be of good commercial quality and as herein specified, and shall conform to the provisions of the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder. All packaging and packing materials, including all components thereof, shall impart no odor or flavor to the product.

3.2 Packaging.- The product shall be packaged in accordance with level A, B, or C, as specified (see 6.1).

3.2.1 Level A.- The ready-to-eat cereals (other than shredded wheat and malted cereal granules) shall be packaged in a rectangular sealed-end carton. Each carton shall be furnished with an interior bag or overwrap constructed from a foil laminated material. The aluminum foil shall be 0.00035-inch thick ± 10 percent (see 4.2.7). The completed bag or overwrap shall consist of one or more laminated structures having a total combined basis weight of not less than 68 pounds per ream (24 x 36-500) (see 4.2.8). The bag or overwrap shall be completely closed by heat sealing.

3.2.1.1 Alternative Level A.- Alternatively, the carton for ready-to-eat cereals (other than shredded wheat and malted cereal granules), shall be furnished with an interior bag or overwrap constructed from a web which shall be completely closed by heat sealing or with a hot melt adhesive. The web shall be constructed from materials which have a water vapor transmission rate of not more than 0.07 grams per 24 hours at 100°F and 90 percent relative humidity for a flat sheet and not more than 0.10 grams per 24 hours for a creased sheet (see 4.2.9) and a tear resistance of not less than 22 grams in the machine direction and not less than 27 grams in the cross machine direction (see 4.2.10).

3.2.2 Level B.- The product shall be packaged in a rectangular, sealed-end carton. Each carton shall be furnished with a sealed, moisture-proof barrier which may be provided as an overwrap, an interior bag or liner, or laminated to the paperboard carton. The carton and barrier material shall be those which are normally used for the product. When applicable, uncooked cereal shall be packaged in paperboard cartons or canisters in accordance with the fill requirements of table I. When specified, (see 6.1), cartons for individual servings shall be of the type which can be used as serving bowls.

3.2.3 Level C.- The product shall be packaged to provide protection against deterioration during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements. When specified (see 6.1), individual servings shall be packaged in plastic bowls.

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TABLE I.- Fill requirements for specified cereals

Product	Net weight
Cereal, Wheat, Uncooked, N-C-201	
Type I	1 lb. 12 oz.
Type IV	1 lb. 6 oz.
Corn Meal (Regular and Degermed), N-C-521	1 lb. 8 oz.
Cereal, Rolled Oats, N-C-195	1 lb. 4 oz.

3.3 Packing.- The product, packaged as specified in 3.2, shall be packed in accordance with level A, B, or C, as specified (see 6.1). Individual servings shall be packed in quantities of 50, 100, 200, 300, or 400 per shipping container, except as noted below. Ten pack assortments shall be packed 24 to a shipping container, and 100 pack assortments shall be packed not more than 4 to a shipping container. Multiple serving cartons shall be packed not more than 48 pounds per shipping container.

3.3.1 Level A.- The product shall be packed into an intermediate box constructed in accordance with style RSC, type CF, class domestic, grade 125 of PPP-B-636, and closed by gluing or taping. The intermediate box shall be placed into a waterproof case liner, fabricated and sealed in accordance with MIL-L-10547 for subsistence items. The filled and sealed intermediate box, inclosed in the waterproof case liner, shall be overpacked in a snug-fitting fiberboard shipping container constructed, closed, and reinforced in accordance with style RSC, V2s of PPP-B-636. If the bottom flaps of the shipping container are closed by stitching, the case liner shall be protected from the stitches by placing a paperboard pad between the stitches and the case liner. The case liner shall be protected from stitches in the manufacturer's joint by placing a strip of pressure-sensitive tape over the stitches.

3.3.2 Level B.- The product, packed in an intermediate fiberboard box and waterproof case liner as specified in 3.3.1, shall be overpacked in a snug-fitting fiberboard shipping container, constructed, closed, and reinforced in accordance with style RSC, V3c, V3s, or V4s, of PPP-B-636. The case liner shall be protected from stitches as specified in 3.3.1.

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3.3.2.1 When specified (see 6.1 and 6.3), the shipping container specified in 3.3.2 shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive, filament-reinforced tape in accordance with the appendix of PPP-B-636, except that one lengthwise band and one girthwise band shall be applied.

3.3.3 Level C.- The product shall be packed in a snug-fitting fiberboard shipping container, constructed and closed in accordance with style RSC, type CF or SF, class domestic, method II closure, of PPP-B-636.

3.3.3.1 When specified (see 6.1 and 6.4), the shipping container shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

3.4 Unit loads.- When specified (see 6.1), the product, packed as specified in 3.3, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified.

3.5 Labeling and marking.-

3.5.1 Labeling of unit containers.- Labeling of unit containers shall be in accordance with the requirements of the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder and in compliance with the requirements of the applicable commodity specification.

3.5.2 Marking of shipping containers and intermediate boxes.- Shipping containers and intermediate boxes shall be marked in accordance with MIL-STD-129.

3.5.3 Unit loads.- Unit loads shall be marked in accordance with MIL-L-35078.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

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4.2.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2 Inspection of the end item.-

4.2.2.1 Examination of the end item for packaging, packing, and marking.- Examination of the end item shall be in accordance with the classification of defects set forth in tables II and III and the inspection levels and acceptable quality levels (AQL's) specified herein. The sample unit shall be one package or one filled and marked shipping container, as applicable. The lot size shall be expressed in terms of packages or shipping container of one type or level for this examination. The AQLs shall be expressed in defects per hundred units (DHU).

		AQL	
Table No.	Inspection level	Major	Total
II	S-3	1.5	6.5
III	S-2	2.5	10

TABLE II.- Examination for packaging

		Category	
Examine	Defect	Major	Minor
Material			
Packaging and packing	Imparts undesirable odor or flavor to product	101	
Unit carton	Not rectangular in form		201
	Type differs from that specified	102	
Labeling	Missing, incorrect, or illegible		202
Unit package	Not clean	103	

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TABLE III.- Examination for packing and marking

Examine	Defect	Category Major	Category Minor
Intermediate container	Contents loose in containers		201
	Less than minimum grade specified (see 3.3.2.1)	101	
	Incorrect closure	102	
Case liner or overwrap	Missing when specified	103	
Shipping container	Contents fit loosely		202
	Count per container not as specified	104	
	Net weight for multiple serving containers more than 48 pounds		203
Marking	Omitted, incomplete, incorrect, illegible	105	

4.2.3 Examination of shipping containers.- When shipping containers are required to be in accordance with PPP-B-636, examination for closure and strapping shall be in accordance with the appendix of that specification.

4.2.4 Examination of unit loads.- Unit loads shall be examined in accordance with the quality assurance provisions of MIL-L-35078.

4.2.5 Examination of waterproof case liner.- When a waterproof case liner is specified, the examination criteria shall be in accordance with MIL-L-10547.

4.2.6 Examination for net weight.- Examination for net weight shall be in accordance with the applicable product specification.

4.2.7 Aluminum foil thickness.- Thickness of aluminum foil shall be determined in accordance with MIL-A-148.

4.2.8 Basis weight of paper.- Basis weight of paper shall be determined in accordance with method 5022 of FED-STD-101.

4.2.9 Water vapor transmission rate.- Water vapor transmission rate shall be determined in accordance with method 3030 of FED-STD-101.

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4.2.10 Tearing resistance.- Tearing resistance shall be determined in accordance with method 2036 of FED-STD-101.

4.3 Testing of interior bag or overwrap.- The interior bag or overwrap specified in 3.2.1 or 3.2.1.1 shall be tested for thickness of aluminum foil and basis weight of paper or for water vapor transmission rate and tearing resistance, as applicable. Test requirements shall be average requirements. The sample unit shall be 1 bag or overwrap, or equivalent in material, and the sample size shall be 5. Results shall be reported to 0.00001 inch for aluminum foil, to 3 significant figures in pound per ream for basis weight, and to 2 significant figures for water vapor transmission rate and for tearing resistance. In lieu of performing tests, conformance to requirements may be determined by examination of pertinent labels, invoices, or other valid documents. Nonconformance shall be cause for rejection of the lot.

5. PREPARATION FOR DELIVERY

(Not applicable).

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Levels of packaging and packing required (see 3.2 and 3.3).
(c) When cartons for individual servings are to be used for serving bowls (see 3.2.2).
(d) When individual servings are to be packaged in disposable plastic bowls (see 3.2.3).
(e) When packing specified in 3.3.2.1 or 3.3.3.1 is required.
(f) Type and class of unit load when unit loading is specified (see 3.4).

6.2 Based on the conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15, NAVSUP PUB 470, AFR 71-6, MCO 4030.140, or DSAR 4145.7, as applicable.

6.3 Packing specified in 3.3.2.1 is intended for transfer at sea operations or specific overseas operations.

6.4 Packing specified in 3.3.3.1 is intended for direct shipment from the supplier to the first receiving activity for immediate use within CONUS.

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Custodians:

Army - GL
Navy - SA
Air Force - 45

Review activities:

Army - MD
Navy - MC

User activity:

Army - SM

Preparing activity:

Army - GL

Project No. 8920-0248

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION Cereals, Breakfast, Prepared (Ready-To-Eat) And Uncooked: Packaging, Packing, Labeling, And Marking Of MIL-C-1470F

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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